

lab_19

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```
library(ggplot2)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(jsonlite)
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
cdc =data.frame(
  year = c(1922L,
           1923L,1924L,1925L,1926L,1927L,1928L,
           1929L,1930L,1931L,1932L,1933L,1934L,1935L,
           1936L,1937L,1938L,1939L,1940L,1941L,
```

```

1942L, 1943L, 1944L, 1945L, 1946L, 1947L, 1948L,
1949L, 1950L, 1951L, 1952L, 1953L, 1954L,
1955L, 1956L, 1957L, 1958L, 1959L, 1960L,
1961L, 1962L, 1963L, 1964L, 1965L, 1966L, 1967L,
1968L, 1969L, 1970L, 1971L, 1972L, 1973L,
1974L, 1975L, 1976L, 1977L, 1978L, 1979L, 1980L,
1981L, 1982L, 1983L, 1984L, 1985L, 1986L,
1987L, 1988L, 1989L, 1990L, 1991L, 1992L, 1993L,
1994L, 1995L, 1996L, 1997L, 1998L, 1999L,
2000L, 2001L, 2002L, 2003L, 2004L, 2005L,
2006L, 2007L, 2008L, 2009L, 2010L, 2011L, 2012L,
2013L, 2014L, 2015L, 2016L, 2017L, 2018L,
2019L, 2020L, 2021L, 2022L, 2023L),
cases = c(107473,
164191, 165418, 152003, 202210, 181411,
161799, 197371, 166914, 172559, 215343, 179135,
265269, 180518, 147237, 214652, 227319, 103188,
183866, 222202, 191383, 191890, 109873,
133792, 109860, 156517, 74715, 69479, 120718,
68687, 45030, 37129, 60886, 62786, 31732, 28295,
32148, 40005, 14809, 11468, 17749, 17135,
13005, 6799, 7717, 9718, 4810, 3285, 4249,
3036, 3287, 1759, 2402, 1738, 1010, 2177, 2063,
1623, 1730, 1248, 1895, 2463, 2276, 3589,
4195, 2823, 3450, 4157, 4570, 2719, 4083, 6586,
4617, 5137, 7796, 6564, 7405, 7298, 7867,
7580, 9771, 11647, 25827, 25616, 15632, 10454,
13278, 16858, 27550, 18719, 48277, 28639,
32971, 20762, 17972, 18975, 15609, 18617, 6124,
2116, 3044, 7063)
)
head(cdc)

```

```

year cases
1 1922 107473
2 1923 164191
3 1924 165418
4 1925 152003
5 1926 202210
6 1927 181411

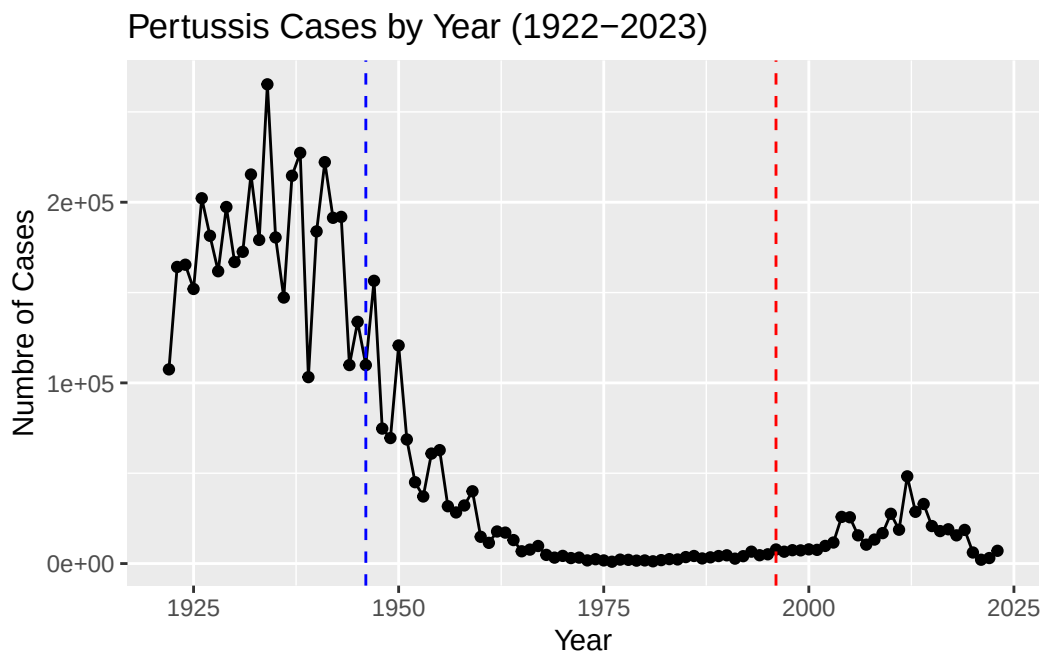
```

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis

case number data to a data frame called `cdc` and use `ggplot` to make a plot of cases numbers over time.

Q2. Using the `ggplot` `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +  
  aes(x=year, y=cases) +  
  geom_point() +  
  geom_line() +  
  geom_vline(xintercept=1946, color="blue", linetype="dashed") +  
  geom_vline(xintercept=1996, color="red", linetype="dashed") +  
  labs(  
    title="Pertussis Cases by Year (1922-2023)",  
    x="Year",  
    y="Numbre of Cases"  
  )
```



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

Bacterial evolution making vaccination less efective. It's also around the time that anti-vax movements started to grow.

```
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
head(subject, 3)
```

```

  subject_id infancy_vac biological_sex ethnicity race
1          1          wP      Female Not Hispanic or Latino White
2          2          wP      Female Not Hispanic or Latino White
3          3          wP      Female      Unknown White
  year_of_birth date_of_boost   dataset
1  1986-01-01   2016-09-12 2020_dataset
2  1968-01-01   2019-01-28 2020_dataset
3  1983-01-01   2016-10-10 2020_dataset
```

```

n_wp = sum(subject$infancy_vac=="wP")
n_ap = sum(subject$infancy_vac=="aP")
n_m = sum(subject$biological_sex=="Female")
n_f = sum(subject$biological_sex=="Male")
```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

aP = 87, wP = 85

Q5. How many Male and Female subjects/patients are in the dataset?

Female = 60, Male = 112

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$biological_sex, subject$race)
```

	American Indian/Alaska Native	Asian	Black or African American
Female	0	32	2
Male	1	12	3

	More Than One Race	Native Hawaiian or Other Pacific Islander
Female	15	1
Male	4	1

	Unknown or Not Reported	White
Female	14	48
Male	7	32

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
subject$age <- today() - ymd(subject$year_of_birth)

ap <- subject %>% filter(infancy_vac == "aP")
print("Age of aP individuals")
```

```
[1] "Age of aP individuals"
```

```
print(round( summary( time_length( ap$age, "years" ) ) ))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

```
wp <- subject %>% filter(infancy_vac == "wP")
print("Age of wP individuals")
```

```
[1] "Age of wP individuals"
```

```
print(round( summary( time_length( wp$age, "years" ) ) ))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	33	35	37	40	58

Q8. Determine the age of all individuals at time of boost?

```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

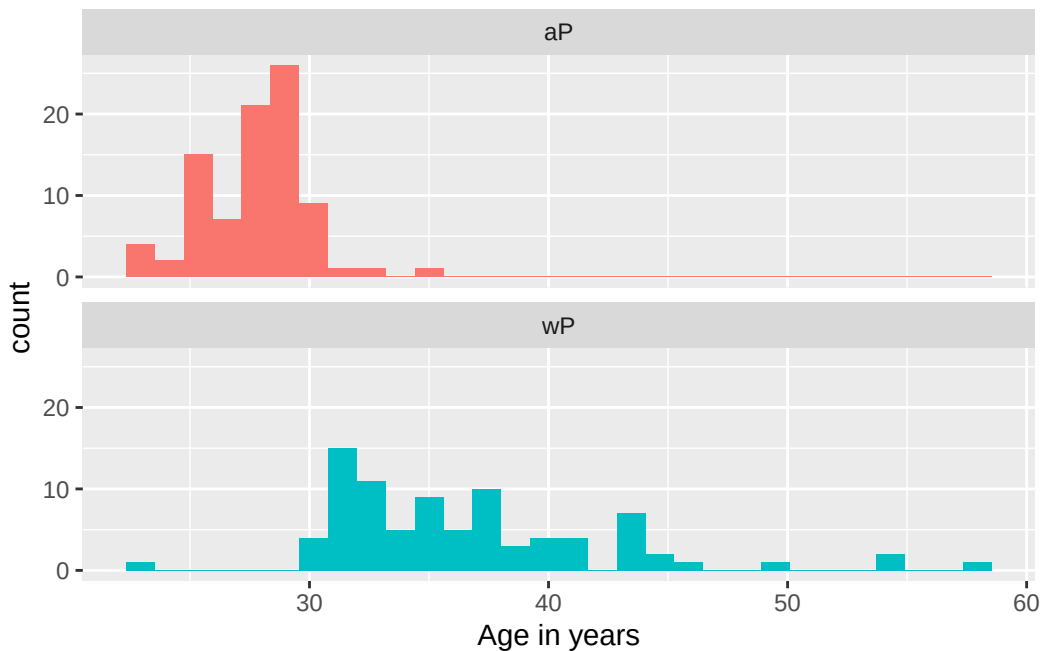
```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

Yes, they are significantly different

```
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



```
# Complete the API URLs...
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)
titer <- read_json("https://www.cmi-pb.org/api/plasma_ab_titer", simplifyVector = TRUE)
```

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
meta <- inner_join(specimen, subject)
```

Joining with ``by = join_by(subject_id)``

```
dim(meta)
```

```
[1] 1503  14
```

```
head(meta)
```

```
specimen_id subject_id actual_day_relative_to_boost
1           1           1                      -3
2           2           1                       1
3           3           1                       3
4           4           1                       7
5           5           1                      11
6           6           1                      32
planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                           0         Blood    1         wP         Female
2                           1         Blood    2         wP         Female
3                           3         Blood    3         wP         Female
4                           7         Blood    4         wP         Female
5                          14         Blood    5         wP         Female
6                          30         Blood    6         wP         Female
ethnicity race year_of_birth date_of_boost dataset
1 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
2 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
3 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
4 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
5 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
6 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
age
1 14585 days
2 14585 days
3 14585 days
4 14585 days
5 14585 days
6 14585 days
```

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```
dim(abdata)
```

```
[1] 52576    21
```

```
head(abdata)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	UG/ML	2.096133	1	-3
2	IU/ML	29.170000	1	-3
3	IU/ML	0.530000	1	-3
4	IU/ML	6.205949	1	-3
5	IU/ML	4.679535	1	-3
6	IU/ML	2.816431	1	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	0	Blood	1	wP	Female
3	0	Blood	1	wP	Female
4	0	Blood	1	wP	Female
5	0	Blood	1	wP	Female
6	0	Blood	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	age
1	14585 days
2	14585 days
3	14585 days
4	14585 days
5	14585 days
6	14585 days

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

```

IgE   IgG  IgG1  IgG2  IgG3  IgG4
6698  5389 10117 10124 10124 10124

```

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```

```

2020_dataset 2021_dataset 2022_dataset 2023_dataset
          31520           8085           7301           5670

```

Each new dataset has a reduction in rows

```

igg <- abdata %>% filter(isotype == "IgG")
head(igg)

```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	IU/ML	0.530000	1	-3
2	IU/ML	6.205949	1	-3
3	IU/ML	4.679535	1	-3
4	IU/ML	0.530000	3	-3
5	IU/ML	6.205949	3	-3
6	IU/ML	4.679535	3	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	0	Blood	1	wP	Female
3	0	Blood	1	wP	Female
4	0	Blood	1	wP	Female

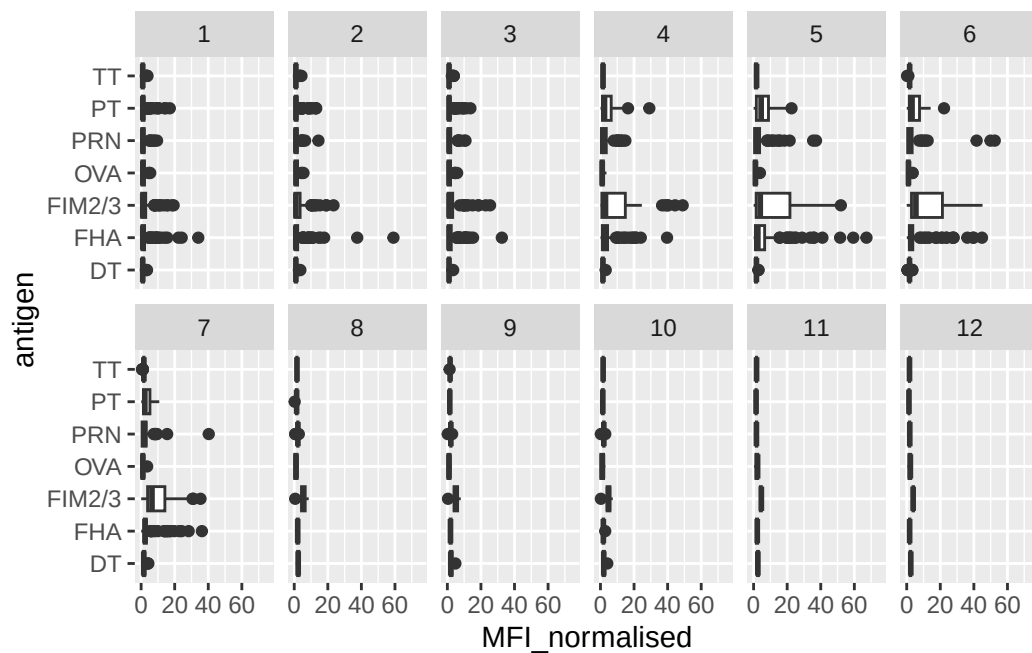
5		0	Blood	1	wP	Female
6		0	Blood	1	wP	Female
	ethnicity	race	year_of_birth	date_of_boost	dataset	
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset	

	age
1	14585 days
2	14585 days
3	14585 days
4	15681 days
5	15681 days
6	15681 days

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

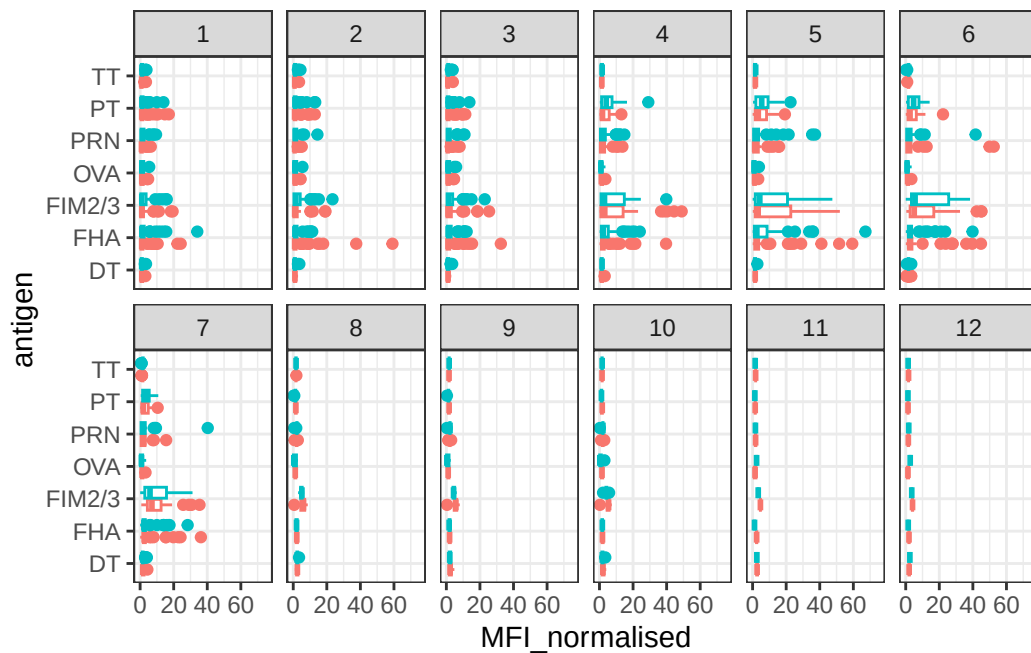
```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



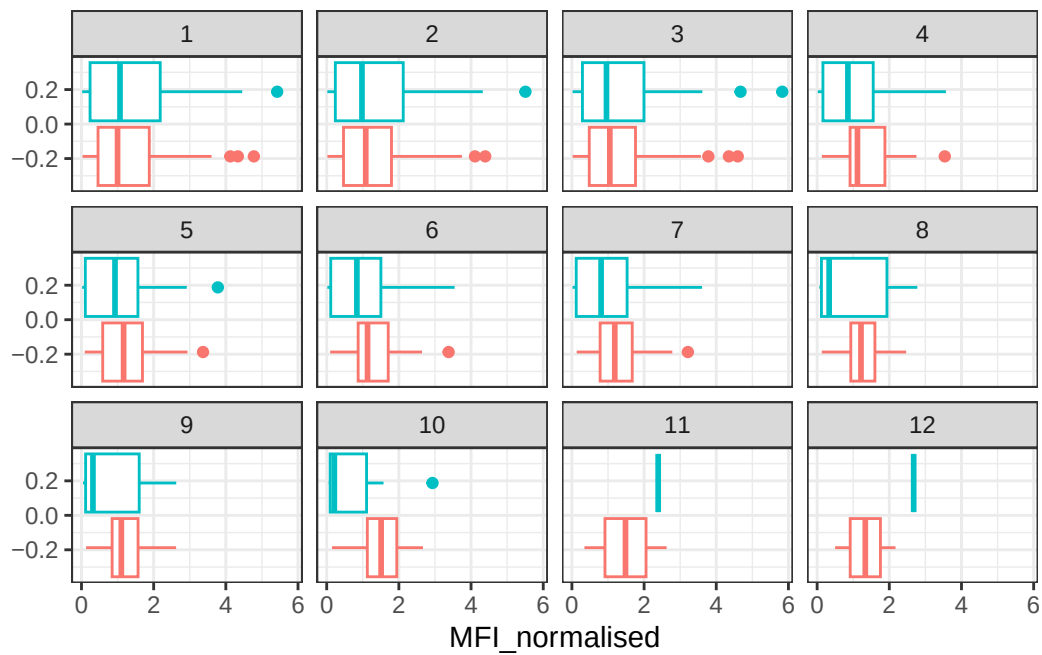
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

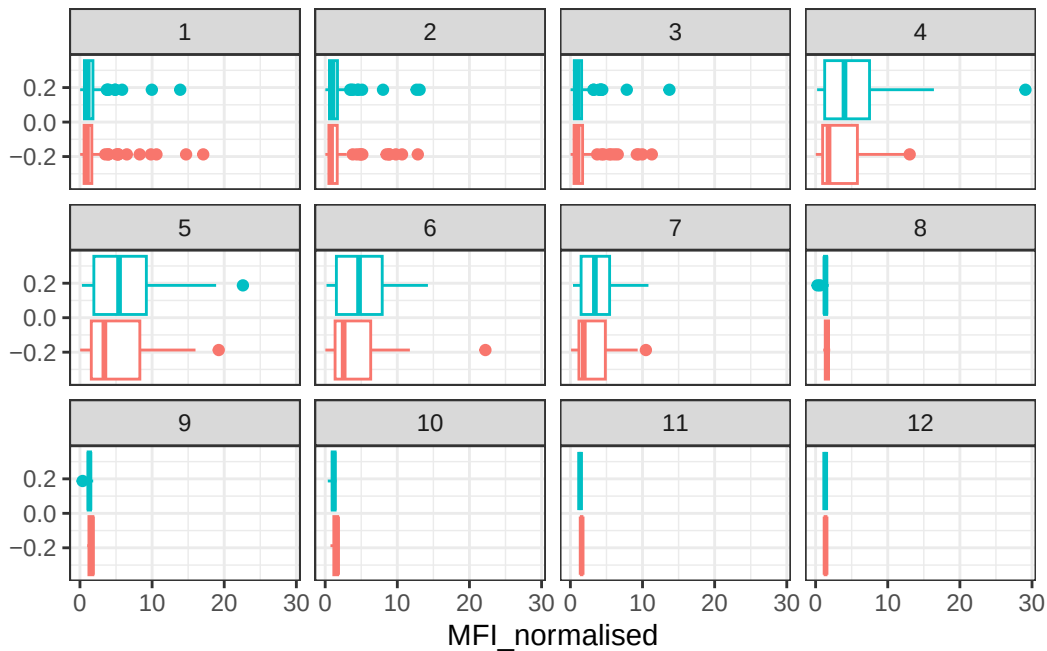


Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a “control” antigen (“OVA”, that is not in our vaccines) and a clear antigen of interest (“PT”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).

```
filter(igg, antigen=="OVA") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



```
filter(igg, antigen=="PT") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

Variance increases and then reduces over time

Q17. Do you see any clear difference in aP vs. wP responses?

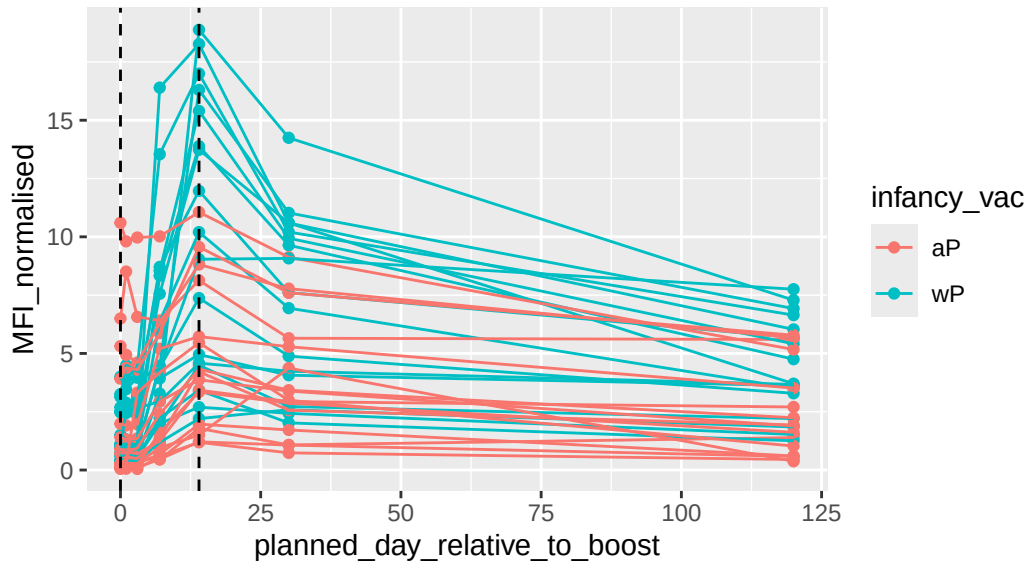
No

```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2021 dataset IgG PT",
         subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)

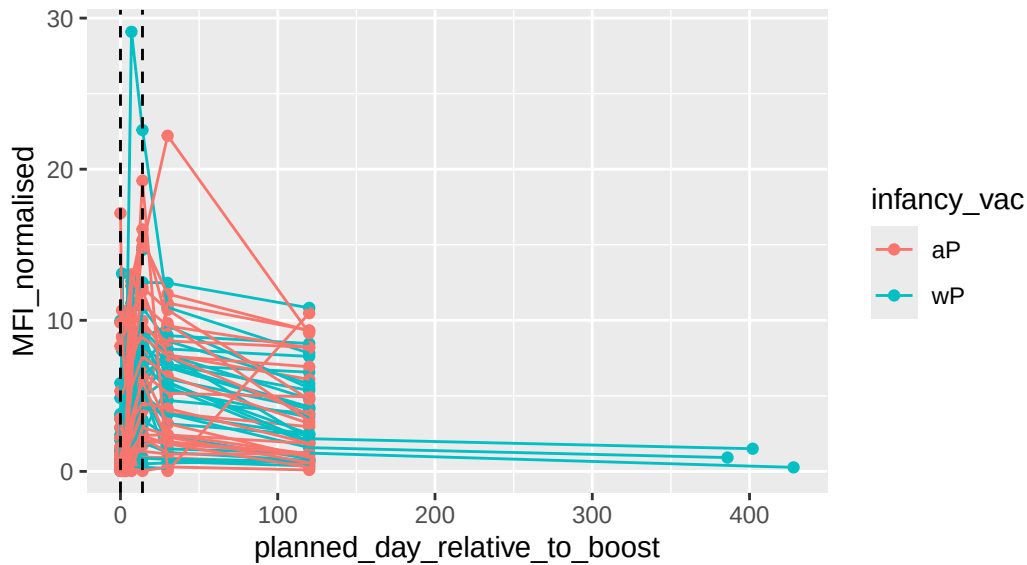


```
abdata.20 <- abdata %>% filter(dataset == "2020_dataset")

abdata.20 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2020 dataset IgG PT",
         subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

2020 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



Q18. Does this trend look similar for the 2020 dataset?

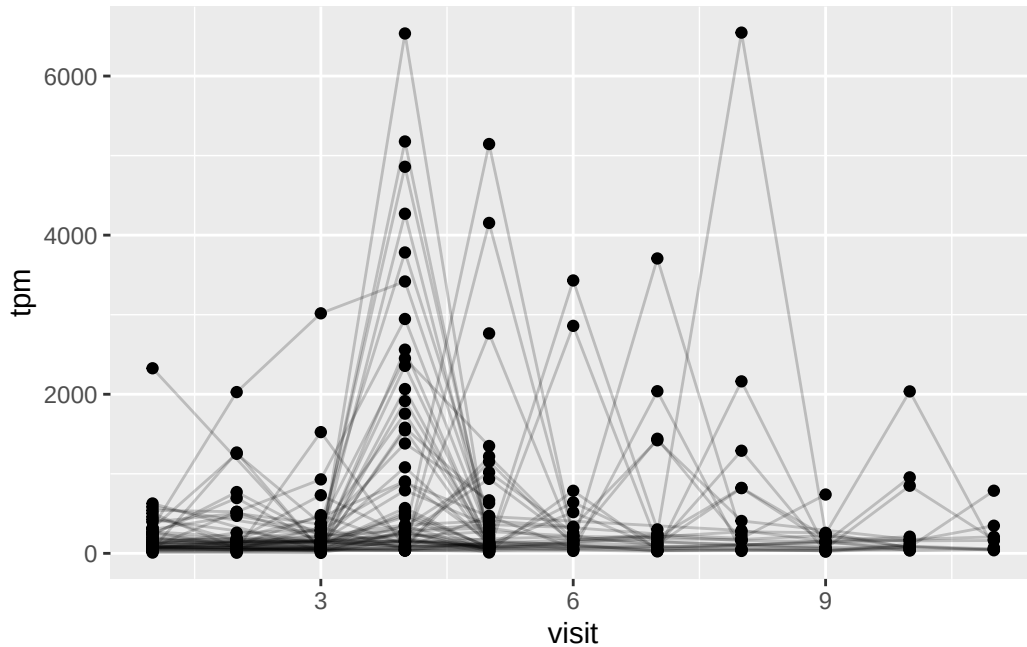
Yes

```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSOG00000211896.7"
rna <- read_json(url, simplifyVector = TRUE)
ssrna <- inner_join(rna, meta)
```

Joining with `by = join\_by(specimen\_id)`

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
ggplot(ssrna) +
  aes(visit, tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```

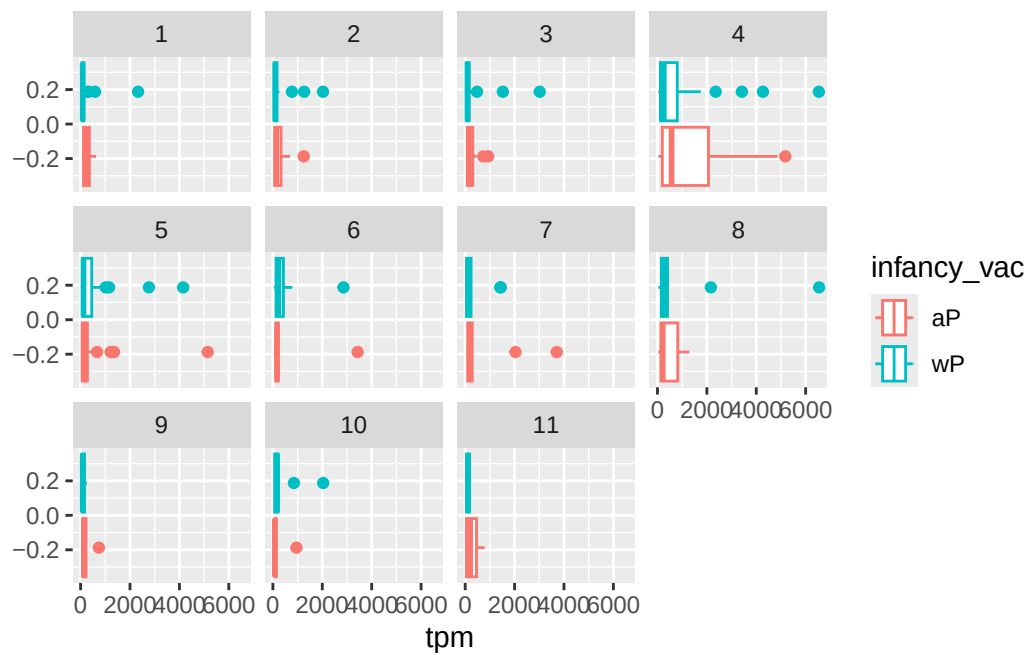
Q20. What do you notice about the expression of this gene (i.e. when is it at it's maximum level)?

It is at the maximum level during the 4th visit

Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

Yes, but it is out of phase in time,

```
ggplot(ssrna) +
  aes(tpm, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(vars(visit))
```



```
ssrna %>%
  filter(visit==4) %>%
  ggplot() +
    aes(tpm, col=infancy_vac) + geom_density() +
    geom_rug()
```

